

Support Vector Machine and Gaussian Process Regression based Modeling for Photovoltaic Power Prediction

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Abstract— Grid integration of Solar energy positively affects energy market due to inexhaustible fuel supply and virtually zero emissions. However, inexhaustible renewable fuel supply is punctuated by the problem of intermittency. Intermittency exacerbates the problem of grid operators to bridge the supply and demand gap. Thus, precise output power forecast of grid interfaced Photovoltaic (PV) systems is required for economic dispatch, market regulation, and stable grid operation. This study compares the statistical models of Gaussian Process Regression (GPR) and Support Vector Machine (SVM) for solar power prediction. The models are trained to predict PV system output power against the backdrop of data recorded for Abbottabad City, Pakistan. Both the models have been trained, validated, and compared with each other for varying irradiance and temperature settings. The results depicted that SVM based modeling excel in solar power prediction with Root Mean Square Error (RMSE) lower than GPR based modeling technique. Performance evaluation of models is conducted with error metrics of RMSE, Mean Absolute Error (MAE), and Mean Square Error (MSE). Moreover, prediction quality is qualified based on residual analysis benchmarked by load line analysis of PV system in Simulink.

Keywords – Gaussian process; Microgrid; Machine learning; Photovoltaic system; Support Vector Machine; Solar Power Prediction

I. INTRODUCTION

The power output of Photovoltaic Generators (PVGs) is intermittent unlike the conventional fossil fuel based power plants [1]. The key phenomena influencing PV systems performance are earth movements and cloud shading. Earth movements are deterministic and hence the PVG output power is calculated precisely during clear sky over various time scales. However, cloud shading is random process. The sudden irradiance variations due to partial shading induce PVG output power variations. The grid operators are concerned for these variations because the abrupt changes in grid interfaced PVG power output will result into grid perturbations [2].

The recent smart grid management requires real time energy production, incorporating demand variations and PVG input variations effectively and efficiently. Thus, forecasting the energy produced by PVGs helps in the effective and secure grid operation. 72 hours ahead

forecasting is of great concern for grid interfaced PVGs [3]. The significance of power forecast is driven by overwhelming grid requirements, ranging from startup time of a conventional power plant to the energy market perspective [4]. The PVG output power depends on solar irradiance that in turns varies with cloud motion and movement of the earth. The grid instability takes place by the abrupt variations of input supply from solar plant to grid. Thus, precise forecast is of key importance. The grid operator notices the difference between predicted and generated power to timely run the backup generators using an accurate forecast. Consequently, the low operation cost along with grid reliability is ensured [5], [6]. Numerous techniques are proposed in the literature to predict PVG power. Few models used exogenous data based on satellite images, sky imager information, and nearby installed PV models, while some other models used nonexogenous information of location [5]. Forecasting span varies from instants to days. Conversely, spatial prediction perspectives are from individual location to a district. Prediction span is categorized as short, medium, and long duration forecast [7]. An Artificial Neural Network (ANN) model based 48 hours ahead PVG output power forecasting technique for Italy was recommended in [8]. They observed that suggested model forecasts PVG power for sunny, partial shading and cloudy day having Root Mean Square Error (RMSE) of 12.5, 24, and 36.9 respectively. Autoregressive Moving Average with exogenous inputs (ARMAX) for PV power prediction was employed in [9]. They established that suggested model improves output precision. A Numerical Weather Prediction (NWP) technique based PVG output power prediction was implemented in [10]. Multivariate Adaptive Regression Splines (MARS) for power prediction of PV system was described in [11]. They compared results with additional techniques as well. They observed that suggested technique achieved better results than other methods for testing and training [11].

The extensive literature exists about comparison of statistical and deterministic techniques for modeling PV systems. Nonlinear behavior of PVG cannot be accurately modeled by using deterministic method. Five parameter deterministic model is compared with Gaussian Process Regression (GPR) based technique and superiority of GPR

over deterministic modeling is established for internal reserve estimation of PVG in [12]. The highlight of this work is to train Support Vector Machine (SVM) and GPR for PV system power prediction. Both the techniques are then compared with each other for accuracy. The technique with low estimated forecasting error is declared as appropriate for output power prediction.

The main contributions of this paper are:

- GPR based statistical modeling of PV system is developed by considering weather conditions of Abbottabad, Pakistan. System is designed for output power prediction of PVG and Maximum Power Point (MPP) estimation in ambient weather.
- SVM technique for power prediction of PVG is also developed for varying temperature and irradiance settings. The model is also formulated for MPP in ambient weather settings.
- Both the statistical models are compared with each other for accurate solar power prediction. Moreover, performance evaluation of models is conducted with error metrics.

The work is structured as: Section II demonstrates statistical techniques of GPR and SVM for output power prediction of PV system. The generally used performance evaluation indicators are also described. Section III presents microgrid architecture, training data description, and comparative analysis of GPR and SVM. Finally, conclusion and future directions are illustrated in Section IV.

II. MATERIALS AND METHODS

Artificial intelligence teaches computer to learn a specific task independently using data without relying on mathematical analysis. Machine learning demonstrates computers to acquire from experience. The algorithms adaptively learn accurately by increasing training data samples. The image and speech recognition, electricity load forecasting, and data mining are numerous applications of machine learning, categorize as supervised and unsupervised learning. Supervised learning predicts future response by training input samples. Regression learning estimates continuous response, while Classification learning estimates categorical output [13]. GPR and SVM techniques of supervised learning is used in this paper for PVG output power prediction.

A. Gaussian Process Regression

In this section, GPR method is summarized for predicting dynamic system of PVG. The regression framework takes training sets $D = \{(x_j, y_j) | j = 1, \dots, n\}$ as input, where n , x , and y presents observations, covariates, and response respectively. Trained GPR forecasts y_{new} for an input x_{new} . The model is represented in (1).

$$y = x^T \beta + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2), \quad (1)$$

where σ^2 denotes error variance and β are coefficients calculated from input data.

Gaussian Process (GP) is definite random variables count having collective Gaussian distribution. GPR model converts (1) by incorporating latent variables $f(x_j), j = 1, 2, \dots, 515$,

from GP and basis functions, h [14]. The $f(x_j)$ covariance function assures smooth response. The conversion of (1) is illustrated in (2).

$$y = h(x)^T \beta + f(x), \quad (2)$$

$$f(x) \sim GP(0, k(x, x')),$$

where $k(x, x')$ depicts covariance function. At any time, y is illustrated as (3).

$$P(y_j | f(x_j), x_j) \sim N(y_j | h(x_j)^T \beta + f(x_j), \sigma^2), \quad (3)$$

$$P(y | f, X) \sim N(y | H\beta + f, \sigma^2 I),$$

The remaining model parameters are initial coefficients ($\beta = 1.0855e + 5$), and variance ($\sigma^2 = 72.97e + 3$). Matern 5/2 is chosen as the kernel or covariance function (f), as illustrated in (4).

$$k(x_j, x_k) = \sigma_f^2 \left(1 + \frac{\sqrt{5}r}{\sigma_l} + \frac{5r^2}{3\sigma_l^2} \right) \exp \left(-\frac{\sqrt{5}r}{\sigma_l} \right). \quad (4)$$

where $r = \sqrt{(x_j - x_k)^T (x_j - x_k)}$ is the Euclidean distance between x_i and x_j [14].

B. Support Vector Machine

SVM is a kernel function based nonparametric technique. The technique is created on Structural Risk Minimization (SRM). The SRM based learning algorithms involve control of two factors namely empirical risk and confidence interval value. First term in inequality of (5) is empirical risk and other is confidence interval.

$$R(\alpha_l^k) \leq R_{emp}(\alpha_l^k) + \Phi \left(\frac{l}{h_k} \right). \quad (5)$$

where l/h denotes relation between training samples and Vapnik-Chervonenkis (VC) dimension of machine's functions. SVM keeps empirical risk value fixed or zero and minimize value of confidence interval. Thus, the task is to search for function that results in minimized error for training data samples. SVM implement sets of functions, as depicted in (6).

$$f(x, \alpha, w) = \text{sign} \left(\sum_{i=1}^N \alpha_i K(x, w_i) - b \right). \quad (6)$$

where N is an integer, scalars in α_i are $i = 1, \dots, N$, while vectors in w_i are $i = 1, \dots, N$. $K(x, w)$ is kernel symmetric function [15]. Continuous function of accurate degree can be approximated by using neural network, polynomial or radial basis function kernel type. The kernels generating polynomials, radial basis function, and neural networks to obtain approximating functions are illustrated in (7), (8), and (9) respectively. Thus, learning machine is categorized by only varying kernel $K(x, w)$ in SVM. The best possible results are achieved, if vectors w_i in (6) overlap with majority vectors of training datasets.

$$K(x, x_i) = [(x * x^*) + 1]^d. \quad (7)$$

$$K(x, x_i) = K(|x - x_i|). \quad (8)$$

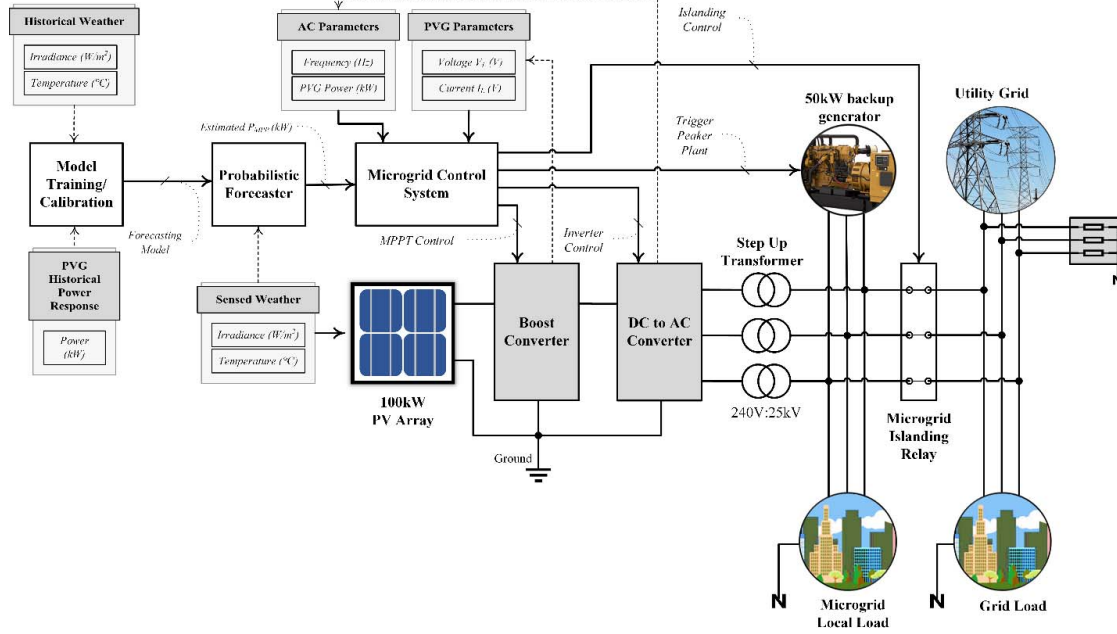


Fig. 1. Schematic of Machine Learning based grid interfaced PV system.

$$K(x, x_i) = S(v(x * x_i) + c). \quad (9)$$

SVM has low complexity and good data fitting feature. SVM performs multiple linear regression employing transformed predictors. The objective is to observe and fine tune three major parameters: precision (ϵ), cost (C) to handle tradeoff concerning model complexity and accuracy, and kernel regulator (γ). The underlying ability of SVM to generalize solutions for non-linear problems is remarkable [15].

C. Evaluation of Model Parameters

Few standard statistical error metrics are employed in this work. The purpose is to evaluate and compare two techniques of GPR and SVM for accurate output power prediction [13]. Mean Absolute Error (MAE) and Mean Biased Error (MBE) are presented in (10) and (11).

$$MBE = \frac{1}{N} \sum_{i=1}^N (P_{m,h} - P_{p,h}). \quad (10)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |P_{m,h} - P_{p,h}|. \quad (11)$$

where $P_{m,h}$ denotes measured power, $P_{p,h}$ represents forecast of prediction method, and N is data samples.

MBE is not a consistent parameter, because data residuals constantly compensate one another. However, MBE presents model's capability for estimation. RMSE is depicted in (12). Meteorology, Economics, and Regression analysis are few fields using RMSE as performance evaluation indicator [13].

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N e_i^2}. \quad (12)$$

III. RESULTS AND DISCUSSION

The comparison of SVM and GPR based PVG modeling is presented here. The model with an appropriate output power prediction is qualified as better technique for opted climate conditions. RMSE, MBE, and MSE serve as prediction quality parameters. Moreover, model reliability is established through visual analysis of residual plots of respective modeling technique.

A. Microgrid Architecture

The system under observation is a microgrid model consisted upon 100kW PVG and 50kW backup diesel generator connected to a utility grid, as depicted in Fig. 1. The diesel generator serves as a peaker plant to balance supply demand variance, after the PVG's internal power reserve is exhausted. Unbalance of supply and demand is reflected by the grid frequency deviation. The power demand greater than supply results in grid frequency rise from nominal 60Hz, and vice versa. Microgrid Control System (MCS) is a centralized microgrid control architecture, that supplies PWM for suboptimal MPP operation[12]. The suboptimal MPP is suggested by either the SVM or GPR model, and is fine-tuned based on the unbalance grid frequency. MCS also generates inverter control signals to produce three phase grid interfaced 240V (RMS) voltages. During utility grid faults, MCS isolates the microgrid by connecting microgrid islanding relay. Finally, in the event of the PVG saturation at 95% of MPP operation, MCS activates secondary frequency protection layer by triggering the peaker plant.

B. Site Description and Data

The training data for both statistical models are yearly irradiance and temperature recorded for Abbottabad, Pakistan. During Autumn and Spring, temperature changes from mild to warm. However, hot weather is experienced in middle of year with cool to mild conditions in Winter time. Heavy monsoon season in July through September is followed by sparse snowfall in the months of December and January [16]. Both GPR and SVM is trained for the climate data of Abbottabad, Pakistan.

The irradiance data illustrated in Fig. 2 as scatter plot is recorded annually in HOMER software. Hours, days, and irradiance are depicted on the x, y, and z axis respectively. Fig. 3 presents average temperature per month during 2009-2017 [17]. Peak temperature is observed in the mid of every year. The relation of maximum output power, temperature, and irradiance is represented in Fig. 4. The collected data of irradiance, temperature, and computed maximum power are used to train GPR Matern 5/2 and SVM technique. Maximum power output is predicted using trained model for ambient irradiance and temperature settings.

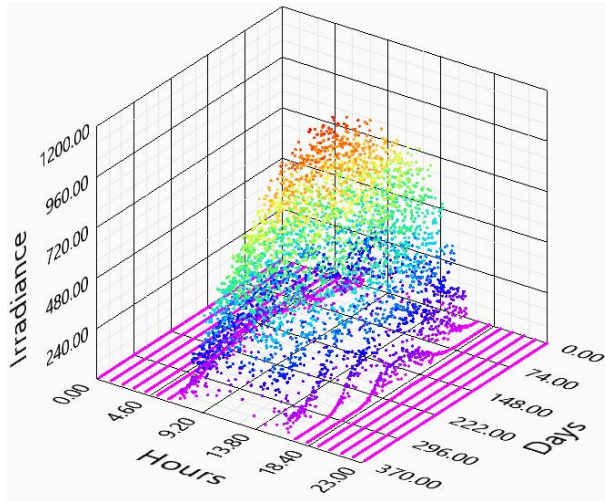


Fig. 2. Photovoltaic annual irradiance model of Abbottabad, Pakistan.

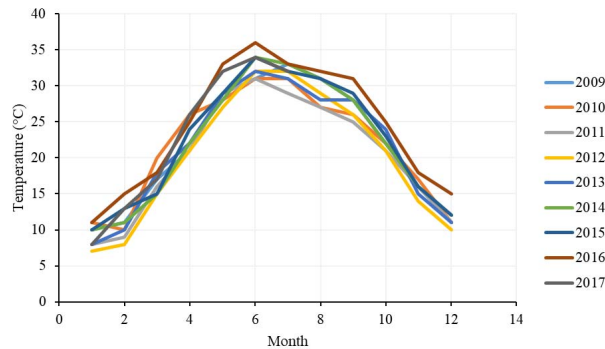


Fig. 3. Average temperature of Abbottabad, Pakistan observed per month in 2009-2017.

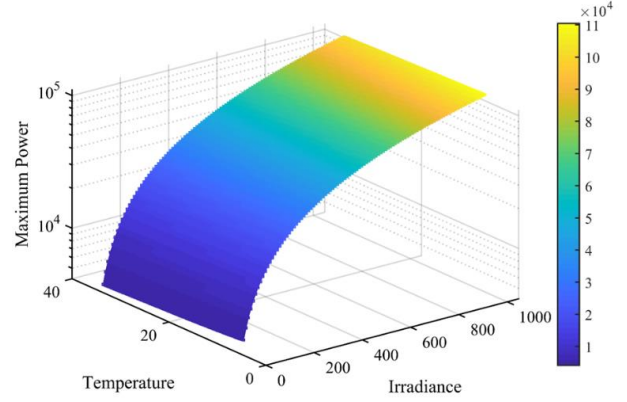


Fig. 4. Training data of Irradiance, Maximum power, and Temperature for GPR and SVM based modeling.

C. Model Training

SVM and GPR models are trained for data obtained from the load line analysis of a 100kW PVG. The climate conditions of Abbottabad, Pakistan are tabulated as training inputs or predictors. The maximum power of PVG obtained from load line study in Simulink is tabulated as training output or response. The 515 predictor samples and respective response observations are fed into Statistics and Machine Learning Toolbox of MATLAB. Model for training are SVM and GPR Matern 5/2 model. The models are anticipated to forecast maximum power output as anticipated by the respective irradiance and temperature, learnt from training. Predicted versus Actual plot in Fig. 5. compares SVM and GPR responses with true response. The technique of best prediction quality is expected to lie flatly on the diagonal line, represented by the true response. Fig. 5. depicts a close matching of the two modeling techniques.

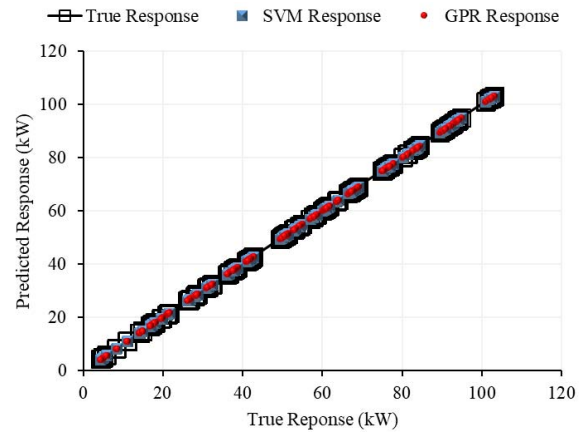


Fig. 5. Response plot comparing SVM and GPR response with true response.

Fig. 6. and Fig. 7. represent individual impact of input predictors of irradiance and temperature on the predicted response, that is PVG output power of two modeling techniques. The individual predictor impact of the two techniques is also closely matched, as anticipated from the Predicted versus Actual plot in Fig. 5.

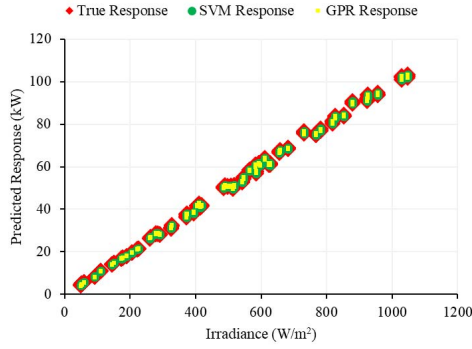


Fig. 6. Irradiance impact on plot comparing SVM and GPR response with true response.

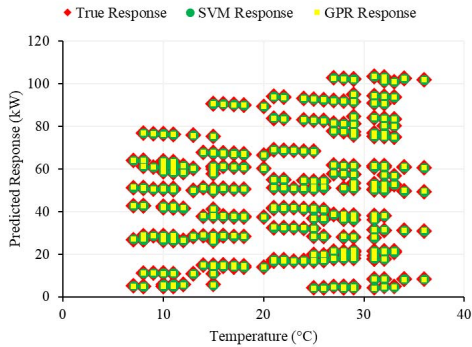


Fig. 7. Temperature impact on plot comparing SVM and GPR response with true response.

Residual plot is another helpful tool to minutely compare the prediction quality of any model. Residual plot displays the error between the predicted and the true response. The model with residuals concentrated symmetrically across the x-axis is expected to perform better. The residuals of SVM are more symmetrically concentrated along the x-axis, as depicted in Fig. 8. GPR offers inferior prediction quality for extremely low irradiance cases as represented by an outlier in Fig. 8. Conversely, SVM offers better prediction quality for all irradiance cases, as evident from x-axis concentrated residuals in Fig. 8.

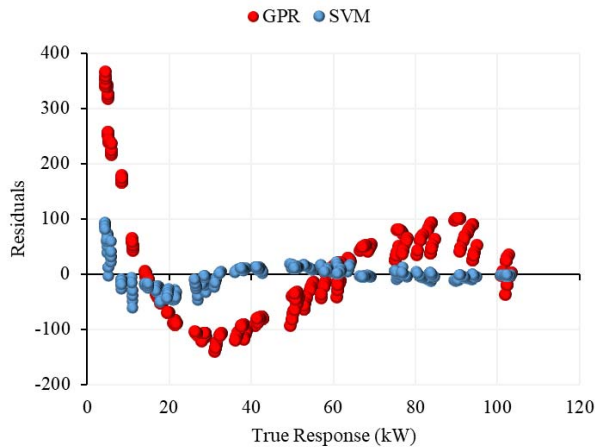


Fig. 8. Residual plot comparing deviation of predicted response from true response for GPR and SVM.

The superiority of SVM is further cemented by comparing performance indices of two techniques, as highlighted in Table I. The indicators opted for evaluation are MSE, RMSE, and MAE. It is clear from Table I that SVM technique estimates nonlinear behavior of PV system more appropriately than GPR and thus accurately predicts output power.

TABLE I. Performance indicators comparison of GPR and SVM.

Performance Indicators	Gaussian Process Regression	Support Vector Machine
Root Mean Square Error (RMSE)	102.32	22.31
Mean Absolute Error (MAE)	76.21	15.43
Mean Square Error (MSE)	10468.72	497.93
Training Time (s)	19.97	16.87

IV. CONCLUSION AND FUTURE WORK

Numerous models for PV system output power prediction are proposed and tested lately. In this work, GPR and SVM based statistical model is trained for output power forecast of PV system. The techniques have been trained, tested, and validated for varying temperature and irradiance conditions of Abbottabad, Pakistan. The performance of both statistical techniques is evaluated using standard statistical error metrics. It is observed that RMSE for GPR is 102.32, while that of SVM is 22.31. Thus, SVM based trained model is better than GPR for proposed scenario of power prediction.

In future, extended microgrid incorporating multiple sources will be simulated. Hardware validation of the statistical techniques will be carried out for testing purpose.

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